

Recall — ratio

What a ratio is, how it differs from a fraction, and how it simplifies.

LESSON 41

1 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 12 ENTRY

LESSON CLOCK

10'

12'

12'

6'

What a ratio says	10 min
Simplifying	12 min
Ratio against fraction	12 min
Reading a ratio back	6 min

LEARNING OBJECTIVES

- Write a ratio from a description and simplify it.
- Convert units before comparing.
- Say what each number of a ratio counts.
- Distinguish a ratio from a fraction of the whole.

TEXTBOOK

National	pp. 109–112
Cambridge	Stage 7, pp. 118–122
Workbook	Exercise 12.1

01

Explanation

What a ratio says

A **ratio** compares two quantities of the same kind. In a class of 12 boys and 18 girls the ratio of boys to girls is 12 : 18, which simplifies to 2 : 3.

SITUATION	RATIO	SIMPLIFIED
12 boys, 18 girls	12 : 18	2 : 3
20 cm to 50 cm	20 : 50	2 : 5
4 red, 6 blue, 10 green	4 : 6 : 10	2 : 3 : 5

THE ORDER MATTERS

2 : 3 is boys to girls; 3 : 2 is girls to boys. Writing the ratio the wrong way round reverses every answer that follows.

Simplifying

Divide every part by their highest common factor — exactly as with a fraction. And convert to the same units first.

RATIO	FIRST STEP	SIMPLIFIED
12 : 18	divide by 6	2 : 3
50 cm : 2 m	50 : 200	1 : 4
30 min : 2 h	30 : 120	1 : 4
0.5 : 2	5 : 20	1 : 4
$\frac{1}{2} : \frac{1}{3}$	multiply by 6	3 : 2

DIFFERENT UNITS MAKE A RATIO MEANINGLESS

50 cm : 2 m is not 25 : 1 — it is 1 : 4. Converting first is not optional; it changes the answer completely.

Ratio against fraction

A ratio compares two parts with each other; a fraction compares one part with the whole.

STATEMENT	MEANING
boys to girls = 2 : 3	for every 2 boys there are 3 girls
boys are $\frac{2}{5}$ of the class	2 of every 5 pupils
girls are $\frac{3}{5}$ of the class	the parts add to 5
boys are $\frac{2}{3}$ as many as the girls	the ratio again, as a fraction

ADD THE PARTS TO GET THE WHOLE

A ratio of 2 : 3 has 5 parts altogether. Every fraction-of-the-whole question starts by adding the parts.

Reading a ratio back

RATIO	TOTAL PARTS	IF THE TOTAL IS 30
2 : 3	5	12 and 18
1 : 4	5	6 and 24
1 : 2 : 3	6	5, 10 and 15
3 : 7	10	9 and 21

ONE PART FIRST, THEN MULTIPLY OUT

Total ÷ number of parts gives one part; each share is that times its own number. Sharing in a ratio, four lessons from now, is exactly this table read forwards.

02 Worked examples

EXAMPLE 1 Simplify the ratio 50 cm : 2 m.

- 1 Convert to the same unit: 50 cm : 200 cm.
- 2 Divide both by 50.
- 3 = 1 : 4

ANSWER

1 : 4

EXAMPLE 2 A class has 12 boys and 18 girls. Write the ratio of boys to girls and the fraction of the class that is boys.

- ① Ratio: $12 : 18 = 2 : 3$.
- ② Total parts: $2 + 3 = 5$.
- ③ Boys are $\frac{2}{5}$ of the class.

ANSWER

$2 : 3$ and $\frac{2}{5}$

EXAMPLE 3 Simplify $\frac{1}{2} : \frac{1}{3}$.

- ① Multiply both parts by 6.
A ratio is unchanged by that.
- ② $3 : 2$

ANSWER

$3 : 2$

03 Interactive model

Count the boys and girls in the room and write the ratio on the board; the class simplifies its own numbers, which is more convincing than any textbook example.

Ratio, or fraction of the whole?

12 : 18 simplifies to:

2 : 3

3 : 2

6 : 9

1 : 2

50 cm : 2 m simplifies to:

25 : 1

1 : 4

50 : 2

1 : 40

With *12* boys and *18* girls, boys are what fraction of the class?

$\frac{2}{3}$

$\frac{2}{5}$

$\frac{3}{5}$

$\frac{1}{2}$

A ratio of 2 : 3 has how many parts?

2

3

5

6

2 : 3 and 3 : 2 are:

the same

different

both wrong

equal to $\frac{2}{3}$

$\frac{1}{2} : \frac{1}{3}$ simplifies to:

1 : 1

3 : 2

2 : 3

1 : 6

30 shared in $1 : 4$ gives:

6 and 24

10 and 20

5 and 25

15 and 15

30 min : 2 h simplifies to:

15 : 1

1 : 4

30 : 2

1 : 2

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
Part	Ulush	Часть
To simplify a ratio	Nisbatni qisqartirish	Сократить отношение
Equivalent ratios	Teng nisbatlar	Равные отношения
Order of the terms	Hadlar tartibi	Порядок членов
Same units	Bir xil birlik	Одинаковые единицы
Total parts	Jami ulushlar	Всего частей
Comparison	Taqqoslash	Сравнение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A ratio compares:

a part with the whole

two quantities of the same kind

two units

nothing

Before simplifying, the units must be:

different

the same

metric

ignored

$20 : 50$ simplifies to:

$2 : 5$

$5 : 2$

$4 : 10$

$1 : 3$

In $2 : 3$ the first quantity is:

larger

smaller

equal

unknown

A ratio of $1 : 2 : 3$ has:

3 parts

6 parts

5 parts

123 parts

Boys to girls $2 : 3$ means girls are:

$$\frac{2}{5}$$

$$\frac{3}{5}$$

$$\frac{2}{3}$$

$$\frac{1}{2}$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $12 : 18$ ANSWER $2 : 3$ 2. Simplify $20 : 50$ ANSWER $2 : 5$ 3. Simplify $4 : 6 : 10$ ANSWER $2 : 3 : 5$ 4. Simplify $50 \text{ cm} : 2 \text{ m}$ ANSWER $1 : 4$ 5. Simplify $30 \text{ min} : 2 \text{ h}$ ANSWER $1 : 4$ 6. The parts of $2 : 3$

ANSWER 5

7. The parts of $1 : 2 : 3$

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. 12 boys and 18 girls: the ratio

ANSWER 2 : 3

2. And the fraction that is boys

ANSWER $\frac{2}{5}$

3. Simplify 0.5 : 2

ANSWER 1 : 4

4. Simplify $\frac{1}{2} : \frac{1}{3}$

ANSWER 3 : 2

5. 30 shared in 1 : 4

ANSWER 6 and 24

6. 30 shared in 2 : 3

ANSWER 12 and 18

7. 30 shared in 1 : 2 : 3

ANSWER 5, 10, 15

Hard · stretch and reason

7 PROBLEMS

1. Simplify $750\text{ g} : 1.5\text{ kg}$

ANSWER $1 : 2$

2. Simplify $45\text{ s} : 3\text{ min}$

ANSWER $1 : 4$

3. If boys to girls is $3 : 7$, what fraction are girls?

ANSWER $\frac{7}{10}$

4. A ratio $2 : 3$ with the first quantity 14: the second

ANSWER 21

5. Simplify $\frac{2}{3} : \frac{4}{9}$

ANSWER $3 : 2$

6. Why must units match before simplifying?

ANSWER Otherwise the two numbers count different things

7. A ratio of $1 : 1$ means

ANSWER the two quantities are equal

07 Homework

Homework — 5 tasks

Convert to the same unit before you simplify anything.

1. Simplify $16 : 24$, $35 : 49$ and $18 : 27 : 45$.
2. Simplify $80\text{ cm} : 2\text{ m}$ and $20\text{ min} : 1\text{ h}$.
3. A class has 14 boys and 21 girls. Write the ratio and the fraction that is girls.
4. Share 40 in the ratio $3 : 5$.

5. Explain the difference between “boys to girls is 2 : 3” and “boys are $\frac{2}{3}$ of the class”.